

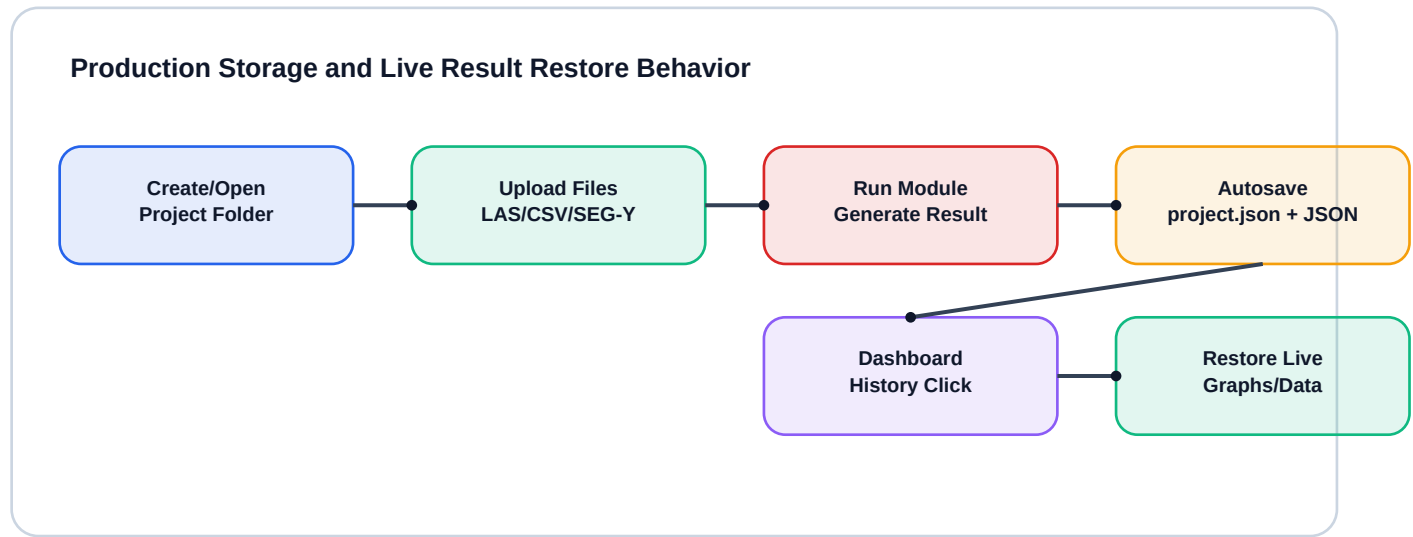
Drake AI Enterprise Platform

Full Project Demo PDF - workflows, modules, result behavior, storage, and downloadable outputs.

Frontend	React 18 + Vite + TypeScript
Backend	FastAPI Python service on port 8002
Project model	Local folder project with project.json, uploads, results, exports, and history
Demo URL	Frontend http://localhost:3000 - API docs http://127.0.0.1:8002/docs

1. Executive Demo Overview

This PDF is a complete demo guide for the current Drake AI project. It starts at login/project creation and ends with module result generation, autosave behavior, dashboard history restore, exports, and production-ready storage behavior.



Area	Demo behavior
Project workspace	Create a folder-backed study, open previous work, and keep uploads/results inside the selected folder.
Result history	Every generated module result writes JSON metadata and appears in Dashboard Result List.
Live result restore	Opening a saved result loads the exact saved payload back into the correct live module view.
Browser storage	Only compact metadata is cached; heavy graphs/results live in project.json and result files.
Production behavior	Direct page reloads hydrate from the folder project file before rendering module pages.

2. Start-to-End User Workflow

- Log in using the Drake AI login screen, then land in the Project Workspace.
- Create a new local folder project or open an existing Drake AI project folder.
- Upload LAS, CSV, SEG-Y, image, report, or table files into project uploads buckets.
- Open a module from the Dashboard and run the desired workflow.
- Generate a result; the app saves the result JSON, project history, export metadata, and live-view state.
- Close the app or switch projects. When the same project opens again, the same project-specific live graphs/data are restored.
- Click a row in Dashboard Result List to reopen that exact result in the proper live module view.

Folder item	Purpose
project.json	Project metadata, dashboard state, module live views, result/export lists, and history.
uploads/las, uploads/tables, uploads/seismic	Copied project input files used by modules.
results/petrophysics, results/ccus, results/geothermal	Saved result JSON files by module domain.
results//exports	CSV, Excel, LAS, image, or other exported files.

3. Dashboard, Project Files, and History

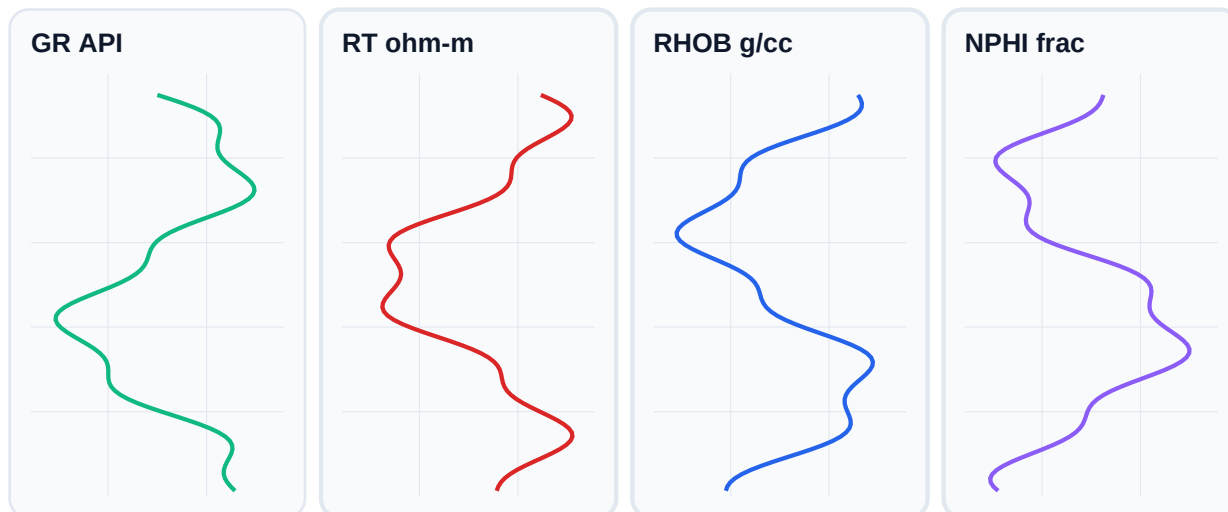
The Dashboard is the operational center. It displays uploaded project files, generated result history, exports, project history, and module launch buttons. Result rows are clickable and restore the selected result into the correct module.

Dashboard section	Demo content	Behavior
Uploaded File List	LAS, CSV, SEG-Y, images, reports	Lists project inputs copied into local project storage.
Result List	JSON result snapshots	Click row to restore exact saved result live view.
Export List	CSV/Excel/LAS/download artifacts	Tracks generated downloadable files.
Project History	Open, upload, result, export events	Auditable project timeline.

4. Petrophysics Demo Results

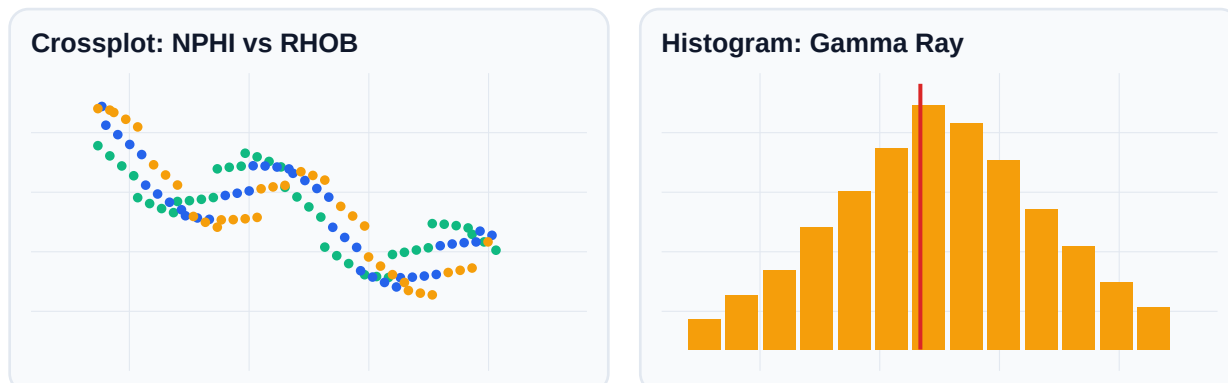
The petrophysics workflow starts with a LAS file, then drives log visualization, missing-log prediction, facies classification, formation tops, AI parameter prediction, uncertainty, crossplots, histograms, and auto-splicing.

Demo LAS Log Visualization - Multi-track View



Depth increases downward - demo curves represent LAS visualization behavior.

Crossplot and Histogram Demo Results



5. Facies, Formation Tops, and Uncertainty

Standalone toolbox modules preserve their own folder-backed live state. A user can run classification or formation-top detection, close the project, and later reopen the exact graph/table state.



Module	Inputs	Outputs
AI Facies Classification	LAS or table log file, depth column, selected features, algorithm	Classified rows, facies strip, CSV export, JSON result
AI Formation Tops	Well log file, optional tops file/manual tops, sensitivity controls	Detected tops, annotated plot, CSV export, JSON result
AI Uncertainty	Active prediction-ready LAS and target settings	P10/P50/P90 envelopes for porosity and saturation

6. Seismic, Production, CCUS, and Geothermal Results

The integrated modules share the same project-first behavior: uploaded files are copied into the project, result JSON is written under the module results folder, and current live views are restored from project.json.

Seismic, Production, CCUS, and Geothermal Results



7. Module-by-Module Demo Matrix

Module	Demo workflow	Saved result behavior
Platform and Projects	Create or open local folder projects, register project.json, upload project files, and continue previous live work.	Autosaves project history, result metadata, and live view state where a live result is generated.
Dashboard	Browse uploaded files, generated results, exports, project history, and open saved results back into live modules.	Autosaves project history, result metadata, and live view state where a live result is generated.
Log Visualization	Upload LAS, inspect well metadata, select curves, render multi-track depth plots, histogram, and crossplot tabs.	Autosaves project history, result metadata, and live view state where a live result is generated.
Missing Log Prediction	Analyze missing curve intervals, select target/features/model, predict gaps, save prediction JSON and live result state.	Autosaves project history, result metadata, and live view state where a live result is generated.
AI Facies Classification	Cluster or classify facies from LAS/table data, render facies tracks and preview classified rows.	Autosaves project history, result metadata, and live view state where a live result is generated.
AI Formation Tops	Detect formation tops from log response changes or uploaded tops/manual tops, then export results.	Autosaves project history, result metadata, and live view state where a live result is generated.
AI Parameter Prediction	Calculate Vsh, porosity, water saturation, lithology, confidence, and preview rows from active LAS.	Autosaves project history, result metadata, and live view state where a live result is generated.
AI Uncertainty	Generate P10/P50/P90 envelopes for porosity and saturation from prediction-ready LAS sessions.	Autosaves project history, result metadata, and live view state where a live result is generated.
Auto Splicer	Validate multiple LAS intervals, splice into a merged LAS, preview output tracks, and download LAS/provenance.	Autosaves project history, result metadata, and live view state where a live result is generated.
Crossplot	Create interactive X/Y scatter plots with optional color-by curve and depth filters.	Autosaves project history, result metadata, and live view state where a live result is generated.
Histogram	Create distribution plots, KDE, statistics, percentiles, quality score, and downloadable figure views.	Autosaves project history, result metadata, and live view state where a live result is generated.
Seismic Frequency Enhancer	Upload SEG-Y/SEG-Y/NPY, choose view/workflow/frequency controls, generate enhanced seismic panels.	Autosaves project history, result metadata, and live view state where a live result is generated.
Production Intelligence	Analyze production CSV/demo data, show optimization candidates, KPIs, and operating recommendations.	Autosaves project history, result metadata, and live view state where a live result is generated.
CCUS Screening	Map LAS curves, calculate reservoir-seal candidates, rank zones, show log panels, and export screening tables.	Autosaves project history, result metadata, and live view state where a live result is generated.
Geothermal Screening	Analyze LAS temperature and reservoir quality, detect hot zones, heat-flow map, and export results.	Autosaves project history, result metadata, and live view state where a live result is generated.
Digitizer SLM/GPT and OCR	Prepare document/image digitization workflows for extraction, review, assistant interaction, and export.	Autosaves project history, result metadata, and live view state where a live result is generated.
Reports and Settings	Provide reporting/export surfaces and workspace preferences without changing module result state.	Autosaves project history, result metadata, and live view state where a live result is generated.

8. Result Restore and Download Behavior

User action	Expected behavior
Generate a module result	Result payload writes to results/*.json and current live view writes to project.json.
Open another project	That project loads only its own saved files, result list, and live views.
Return to previous project	Previous live graphs and tables appear again after folder-backed hydration.
Click Dashboard Result List row	The app reads exact saved result JSON and opens the correct module route.
Download export	Export file is copied to project results exports folder and recorded in Export List.
Browser storage quota	Only compact metadata is cached; large result data remains inside the project folder.

Production acceptance checklist

- Open project folder permission is requested when needed by the browser File System Access API.
- project.json remains the source of truth for live module state and result history.
- Saved result payloads are kept as JSON for reproducible live views.
- Large graph payloads are not pushed into localStorage.
- All UI pages hydrate from the project folder before rendering live module state.

9. Demo Script for Stakeholders

Step	Presenter action	Talking point
1	Create or open a project folder	Drake AI runs as a project-first workspace.
2	Upload LAS/CSV/SEG-Y files	Inputs are copied to structured project storage.
3	Open Log Visualization and run plot	Multi-track graphs are generated from the active LAS.
4	Run Crossplot or Histogram	Result visualizations are saved and appear in Dashboard Result List.
5	Switch project and reopen the first project	Live graphs/data are project-specific and restored from project.json.
6	Click a saved result row	The exact historical result is reopened in the live module page.
7	Export/download outputs	Artifacts are tracked in Export List and stored inside the project folder.

End of demo PDF. This document is generated from the current Drake AI workspace and is intended as a downloadable stakeholder walkthrough.